

HARDIK SRIVASTAVA

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Portfolio | LinkedIn | LeetCode | ResearchGate | Google Scholar | GitHub

SKILLS

ML/Research Sequence/Probabilistic Modeling, Semi-Supervised Learning, Graph-ML, NLP, Ranking, Transformers, xAI
CS/Tooling C++, Python, SQL, PyTorch, TensorFlow, Caffe, Sklearn, Spark, Kafka, W&B, HuggingFace, Optuna
Infra/MLOps AWS, Docker, Kubernetes, Flask, MLFlow, NVIDIA Triton, Feature Stores, Data Version Control, Jenkins

EDUCATION

University of Washington Seattle, WA
Master of Science in Data Science Sept 2025 - Present

SRM Institute of Science and Technology Chennai, TN
B.Tech in Computer Science Engineering with Specialization in Big Data Analytics (CGPA - 9.56/10) Jun 2023

EXPERIENCE

Applied Scientist 2 (Prev. Applied Scientist 1) Jun 2023 - Aug 2025
JPMorgan Chase & Co. Hyderabad, TS

- Led **fraud detection** initiatives saving over **\$3B annually** by developing high-recall classifiers using **XGBoost** and **CatBoost** with **multi-objective optimization** and **monotonic constraints** to preserve feature interpretability.
- Researched and deployed semi-supervised graph neural network models on 60M+ transaction nodes using **topology-aware edge sampling** and **dynamic neighborhood aggregation**; boosted fraud-detection precision by 9.2% over 2M trans/day.
- Designed and productionized a CLIP-style **Contrastive Pre-Trained Transformer** (ViT encoder & BERT-style decoder) for Check OCR, processing 500K+ checks in under 4 hours daily with a **19% boost** in field-level text matching rate.
- Built **Learning-To-Rank** entity matching models with **Pairwise Ranking Loss**, outperforming the **BM25/TF-IDF** baselines on a 10M+ record dataset. Mitigated **62% false positives** and saved **700+ manual review hours** daily.
- Developed **Data Engineering** pipelines for high-volume transactions using PySpark. Performed **Feature Selection** using **Decision Trees** on **3x more features** than legacy models. Optimal features boosted **Fraud-Capture Rate** by **14%**.
- Collaborated with **JPMC AI Research** on an **internal whitepaper** on **xAI techniques** (SHAP, monotonicity constraints and surrogate tree approximators); **mentored 2 interns** on **cross-lingual NER** and **drift detection**.

Research Intern Jun 2022 - Sept 2022
McGill University - *Mitacs Scholar* Montreal, QC

- Developed a **Vocabulary Expansion** model for **AAC research** using Affinity Propagation and **SWOW** embeddings. Baseline comparisons (**COCA**) revealed a **4.41 unit mAP gain** and a **21% decrease** in OOV words for test cues.
- Developed **RoBERTa-based sentence embeddings** to capture word-context in image captions. Achieved **Krippendorff's** alpha reliability score of **0.82**; deployed and productionized in **Click AAC** mobile application.

Machine Learning Engineer - Intern Jan 2022 - Jun 2022
Samsung Electronics Bangalore, KN

- Optimized **word-context relationships** for Bixby search using bipartite graphs, reducing **false-positive** preds by **12%**. Matched **FP32** perf for INT8 inference using quantization-aware training for word-disambiguation task with **2.9% F1 gain**.

PROJECTS

Auto-Explainability and Evaluation of LLMs Developed a framework for Transformer explainability by visualizing prediction attributions using Integrated Gradients in TensorFlow. Integrated with MLFlow Evaluation API for auto-evaluation.

No-Code Algo-Trading Engine Developed a domain-specific language with Mean Reversion strategies to create & execute trades. Added a risk-assessment model using Backpropagation, reducing risk by 15.2% compared to open-sourced models.

PUBLICATIONS

- [Srivastava, H.](#) (2023). Multi-Modal Sentiment Analysis Using Text and Audio for Customer Support Centers
- [Srivastava, H.](#) (2023). Neural Text Style Transfer with Custom Language Styles for Personalized Communication Systems
- [Srivastava, H.](#) (2021). Using NLP Techniques for Enhancing Augmentative and Alternative Communication Applications

ACCOMPLISHMENTS

- Certifications** — AWS Certified Cloud Practitioner, AWS Certified Machine Learning Specialist
- 1st Position** — JPMC ScaleUp (2024), JPMC AWS DeepRacer Regionals (2023), Smart India Hackathon Nationals (2023)